JAMA Ophthalmology | Original Investigation | JAMA EYE ON AI

Smartphone-Based 3D Ocular Reconstruction for Automated Quantitative Blepharoptosis Evaluation

Yahan Yang, MD, PhD; Yuxuan Han, PhD; Ruixin Wang, MD, PhD; Sinting Ling, MS; Junfeng Lyu, PhD; Chengkai Li, PhD; Lanqin Zhao, MSc; Qianni Wu, MD, PhD; Jingyi Wen, PhD; Tingxin Cui, MD; You Li, MD, PhD; Yunxi Lai, MD; Danqi Zeng, MD; Xinwei Chen, MS; Yongxin Zhang, MD; Lin Ye, MD, PhD; Weihua Yang, MD; Xiuyun Gao, MSc; Jianmin Hu, MD, PhD; Can Gong, MS; Jiemin Zou, MD, PhD; Jie Meng, MD; Xingyi Li, MD; Xuanwei Liang, MD, PhD; Danping Huang, MD, PhD; Patrick Yu-Wai-Man, MD, PhD; Feng Xu, PhD; Haotian Lin, MD, PhD

IMPORTANCE External ocular examination is central to ptosis evaluation, yet it often relies on manual, experience-dependent measurements with variable accuracy and reproducibility.

OBJECTIVE To develop and prospectively evaluate the Digital Rulers of Ptosis (DRP), a smartphone-based 3-dimensional (3D) reconstruction system for automated quantitative external ocular assessment in blepharoptosis, including marginal reflex distance 1 (MRD1) and levator muscle function (LF), and clinical classification.

DESIGN, SETTING, AND PARTICIPANTS This diagnostic/prognostic study was conducted prospectively across 5 hospitals in 2025. Participants 3 years and older underwent smartphone video capture and same-day oculoplastic specialist assessment as the reference standard. Participants undergoing surgery received repeat specialist measurements of MRD1 and LF at 3 months postoperatively to assess recurrence.

EXPOSURE DRP uses a 10-second smartphone video for 3D reconstruction with metric unit conversion and real-time quality control.

MAIN OUTCOMES AND MEASURES Agreement between DRP and specialist measurements was assessed by mean absolute error (MAE) and intraclass correlation coefficient. Diagnostic performance was assessed by area under the receiver operating characteristic curve (AUC), sensitivity, specificity, positive predictive value, negative predictive value. For patients who underwent surgery, 3-month postoperative recurrence was assessed.

RESULTS Among 952 participants (1904 eyes), the mean age was 27.9 years (range, 3-86 years); 493 participants were female (51.8%) and 459 were male (48.2%). A total of 1862 eyes were included in MRD1 analysis and 1830 eyes in LF analysis after quality control. DRP demonstrated high agreement with specialist measurements for MRD1 (MAE, 0.52 mm; 95% CI, 0.50-0.54 mm) and LF (MAE, 0.95 mm; 95% CI, 0.91-0.98 mm). For diagnosing ptosis, DRP achieved an AUC of 0.97 (95% CI, 0.96-0.97). For identifying surgical approach, DRP achieved an AUC of 0.98 (95% CI, 0.97-0.98). Performance remained consistent across sex, age, and geographic region. Among 206 surgical participants (248 eyes), the 3-month postoperative recurrence rate was 7.3% (95% CI, 4.4%-11.2%; 18/248 eyes), within the lower range of previously reported rates.

CONCLUSIONS AND RELEVANCE This study found that accurate, standardized quantitative blepharoptosis assessment might be supported by commercially available smartphones, demonstrating the potential of smartphone-based 3D reconstruction for external ocular assessment and clinical decision-making. Further studies are needed to evaluate its clinical impact.

JAMA Ophthalmol. doi:10.1001/jamaophthalmol.2026.4120
Published online September 24, 2026.

Multimedia

Supplemental content

Author Affiliations: Author affiliations are listed at the end of this article.

Corresponding Authors: Haotian Lin, MD, PhD, Zhongshan Ophthalmic Center, Sun Yat-sen University, No. 7 Jinsui Rd, Guangzhou 510623, China (linht5@mail.sysu.edu.cn); Feng Xu, PhD, Tsinghua University, Room 1411, Bldg 2, Ziqiang Science and Technology Building, Haidian District, Beijing 100084, China (feng-xu@tsinghua.edu.cn).

Downloaded from jamanetwork.com by Tsinghua University user on 10/27/2026

External ocular examination is important for screening, diagnosis, severity assessment, and surgical planning because many ocular and systemic disorders manifest through abnormalities in eyelid position, ocular alignment, or eye movements.[1-3] This is particularly relevant in blepharoptosis (ptosis), characterized by an abnormally low upper eyelid position, where precise quantification of eyelid position and function is essential for diagnosis and surgical planning. In children, ptosis may obstruct the visual axis and impair visual development[4-6]; in adults, it may indicate underlying neurological or cerebrovascular disorders, and impair quality of life and psychosocial well-being.[7]

Margin reflex distance 1 (MRD1) and levator muscle function (LF) are key measures for ptosis diagnosis and surgical approach selection, and their precision is critical to postoperative prognosis.[3] However, standardized quantitative ptosis assessment remains challenging for nonspecialists. MRD1 and LF are commonly measured manually and are vulnerable to examiner experience, patient cooperation, head position, fixation, and measurement angle, particularly in young children. Inaccurate preoperative quantification may contribute to suboptimal surgical planning and postoperative recurrence, with reported recurrence rates ranging from 14.1% to 63.2%.[8-13] Beyond in-clinic assessment, standardized quantitative data may support postoperative follow-up, longitudinal childhood monitoring, and monitoring of potential ptosis progression or fluctuation. However, caregivers generally cannot reliably measure MRD1 or LF at home, limiting standardized quantitative data between visits for clinicians. Thus, improving measurement accuracy and accessibility may support screening, follow-up, and surgical planning.

Advances in smartphones and artificial intelligence have accelerated image-based external ocular assessment, but most 2-dimensional (2D) image-based methods provide classification rather than clinically quantification.[14,15] Precise measurement from 2D images remains difficult and often requires specialized equipment or highly standardized capture conditions.[16-18] LF assessment is particularly vulnerable to head movement and measurement-angle bias because it depends on dynamic tracking of eyelid excursion. Additionally, the use of raw facial images raises significant patient privacy concerns, hindering clinical adoption.[19] Three-dimensional (3D) reconstruction may address these limitations by modeling eyelid and eyeball morphology and movement, calibrating relative motion among the head, eyelids, and eyeballs and separating disease-relevant features from identifiable facial information.[20,21] In our prior work, we developed a 3D reconstruction-based "digital mask" to preserve patient privacy and quantify strabismus deviation.[22,23]

In this prospective multicenter study, we developed and evaluated Digital Rulers of Ptosis (DRP), a consumer smartphone-based 3D reconstruction system for automated MRD1 and LF quantification, implemented within a smartphone application for clinical use and patient self-monitoring. We evaluated its agreement with oculoplastic specialist measurements, diagnostic and surgical decision performance, comparison with a 2D deep learning baseline, and 3-month postoperative recurrence.

Key Points

Question Can a smartphone-based 3-dimensional (3D) facial reconstruction system provide accurate automated quantitative assessment of blepharoptosis that might support clinical decision-making?

Findings In this diagnostic/prognostic study, the image-based Digital Rulers of Ptosis (DRP) system showed high agreement with specialist manual measurements for marginal reflex distance 1 and levator function, achieving high accuracy for blepharoptosis diagnosis and surgical strategy classification, with low postoperative recurrence rate.

Meaning These results suggest smartphone-based 3D reconstruction may enable accurate, standardized, and scalable external ocular assessment to support quantitative diagnosis and treatment decisions for blepharoptosis in the clinical practice setting.

Methods

The study protocol was approved by the Ethics Review Committee from Zhongshan Ophthalmic Center. The clinical trial was prospectively registered in the Chinese Clinical Trial Registry (identifier: ChiCTR2500112450). The study followed the Declaration of Helsinki and the Standards for Reporting of Diagnostic Accuracy (STARD) reporting guideline (eTable 1 in Supplement 1).

During the prospective clinical evaluation stage, written informed consent was obtained from all participants or from at least 1 legal guardian, as appropriate. Participants were offered the opportunity to receive 1 free clinical examination. No additional financial compensation or other incentives were provided for study participation.

Research Design

DRP was developed as a smartphone-based intelligent system for ptosis diagnosis. It automatically quantifies MRD1 from a 10-second video, identifies the presence of ptosis, and then quantifies LF to suggest an appropriate surgical type. In this prospective multicenter study across 5 clinical centers, DRP's assessments were compared with measurements by experts in the outpatient clinic setting as the reference standard (eFigure 1 in Supplement 1). For patients who underwent surgery, MRD1 and LF were remeasured by experts at 3 months postoperatively (reference standard) to assess recurrence.

Data Acquisition and Clinical Definitions

The examination procedure aligned with routine clinical practice (**Video**). Each video captured sequential MRD1 and LF assessment, and 3 key frames were extracted for analysis: forward gaze, downgaze, and upgaze. MRD1 was defined as the vertical distance from the central upper eyelid margin to the midpupil in forward gaze. MRD1 less than 4 mm indicated reduced eyelid height or droop from the normal eyelid position, and MRD1 2 mm or less was the prespecified anatomic threshold for ptosis in this diagnostic accuracy study.[24-26]

Downloaded from jamanetwork.com by Tsinghua University user on 10/27/2026

LF was defined as upper eyelid excursion from extreme downgaze to extreme upgaze with frontalis muscle action blocked and was categorized as good (LF ≥7 mm; levator resection or levator advancement), fair (LF 4-<7 mm; levator resection), or severe (LF <4 mm; alternative levator muscle surgery such as frontalis sling may be considered).[27,28] Standardized LF acquisition was supported by protocol-based operator training, an in-app demonstration video, voice guidance, and automated quality control for eyebrow movement during the downgaze-to-upgaze sequence.

DRP Model Construction Based on 3D Reconstruction

Building on our prior digital mask research, we advanced the development of the DRP system.[22] DRP digitizes ptosis features using 3D reconstruction and neural network algorithms,[20] integrating principles of medical examination. The reconstruction pipeline comprises 3 primary stages: facial reconstruction, eyelid reconstruction, and eyeball reconstruction. In each stage, a dedicated deep learning-based detector extracts relevant features (facial landmarks L-face in the first stage, eyelid landmarks L-eyelid and semantic lines S-eyelid in the second stage, and iris landmarks L-iris in the final stage). Following feature acquisition, corresponding model parameters are optimized to fit these features.

While based on our previous digital mask reconstruction pipeline, the method was adapted for ptosis evaluation in several key ways (eMethods in Supplement 1): (1) Faster 3D reconstruction: DRP reconstructs only 3 key frames (forward gaze, downgaze, and upgaze) instead of all video frames as a digital mask, accelerating 3D reconstruction. (2) Unit conversion: iris diameter-based scale conversion was applied to the reconstructed 3D geometry to calculate LF and MRD1 in real-world units. (3) LF metric calculation: the reconstructed 3D faces from the upgaze and downgaze frames are transformed into a canonical head pose. Then, we can calculate the moving distance of the upper eyelid as the LF metric by computing the distance between corresponding face vertices in the 3D space. The correction of head-pose variation before LF calculation is illustrated in eFigure 2 in Supplement 1. (4) MRD1 metric calculation: MRD1 was directly and automatically calculated for each eye from the forward-gaze frame as the 3D distance between the iris center and upper-eyelid vertices.

Model Quality Control Design

To optimize external eye video acquisition, a 3-step quality control module was designed to ensure that collected examination videos were clinically acceptable and computationally analyzable, including successful landmark detection, landmark quality assessment, and examination compliance check (eMethods in Supplement 1), which helped identify videos in which dermatochalasis or other factors obscured key eyelid or iris landmarks. Data passing all these steps proceeded to metric calculation. Otherwise, users received the failure reason and a request for recapture.

Prospective Clinical Evaluation

A prospective clinical evaluation of DRP was conducted at Zhongshan Ophthalmic Center, Sun Yat-sen University, The First Affiliated Hospital of Sun Yat-sen University, The Second Affiliated Hospital of Fujian Medical University, Shenzhen Eye Hospital, and Loudi Central Hospital in 2025. Inclusion criteria were as follows: (1) age at least 3 years, reflecting the population commonly entering standardized hospital-based preoperative ptosis assessment and surgical planning; (2) ability to stably fixate on a target with both eyes; and (3) provision of signed informed consent. Video data had to pass the quality control module, with a frame rate 25 frames per second or more and resolution 720 p or greater. At each center, ptosis examinations and video recordings were performed by 2 trained nonspecialists, while an oculoplastic specialist simultaneously measured MRD1 and LF during the same visit. DRP results were then compared with the specialist's measurements.

Comparison With 2D-Based Methods

To highlight the advantages of our 3D approach, DRP was compared against 2D-based approaches that use convolutional neural networks on red-green-blue (RGB) images as input to output ptosis severity. First, the same method was used to detect facial landmarks. For MRD1, 2D pixel distance was calculated between the iris center and the upper eyelid in the images. In contrast, DRP calculates this distance directly in calibrated 3D space. For LF, we calculated the pixel distance between the upper and lower eyelids in both the upward and downward gaze frames, taking the difference between these 2 distances to approximate the moving distance of the upper eyelid. DRP eliminates the effects of head motion by performing the reconstructed face in a canonical space, and thus we can directly compute the upper eyelid's movement without such approximation.

Statistical Analysis

The sample size was calculated for the primary outcome, the intraclass correlation coefficient (ICC). Using the method of a 1-way random-effects analysis-of-variance model with 2 observations per participant, an expected ICC of 0.885 (from pilot data), and a target 95% CI width of 0.05, a sample size of 294 participants was required. Accounting for a 2% dropout rate, the final target was 301 participants. Calculations were performed using Power Analysis and Sample Size (PASS) software version 16 (NCSS).

DRP performance was evaluated against the reference standard (expert manual measurements) in 2 domains: quantitative and classification analyses. For quantitative assessment, agreement between DRP and expert measurements was evaluated using the mean absolute error (MAE) and ICC based on a 1-way random-effects model. Bland-Altman plots were also generated to visualize the agreement and identify any systematic bias. For classification performance, the area under the receiver operating characteristic curve (AUC), sensitivity, and specificity were calculated. The strength of the linear relationship between DRP and expert measurements was further assessed using Spearman correlation coefficients. All *P* values were 2-sided but not adjusted for multiple analyses. All statistical analyses were performed using R version 4.4.2 (R Foundation),

Downloaded from jamanetwork.com by Tsinghua University user on 10/27/2026

Table. Clinical Characteristics of the Dataset

Characteristics	No./total No. of eyes (%)
Sex	
Male	918/1904 (48.2)
Female	986/1904 (51.8)
Age group, y	
≤6	412/1904 (21.6)
7-59	1278/1904 (67.1)
≥60	214/1904 (11.2)
Institution	
Zhongshan Ophthalmic Centre	1358/1904 (71.4)
Other hospitals	546/1904 (28.6)
Diagnosis	
Normal/other, MRD1 ≥4 mm	880/1862 (47.3)
Droop from normal position, MRD1 2-<4 mm	451/1862 (24.2)
Ptosis, MRD1 ≤2 mm	531/1862 (28.5)
Severity of levator muscle function	
Good, LF ≥7 mm	1215/1830 (66.4)
Fair, LF 4-<7 mm	373/1830 (20.4)
Severe, LF <4 mm	242/1830 (13.2)
Etiology	
Congenital	485/982 (49.3)
Acquired	292/982 (29.7)
Unknown	205/982 (20.9)
Recurrence after surgery	
Preoperative levator muscle function = fair	14/248 (5.6)
Preoperative levator muscle function = severe	4/248 (1.6)

Abbreviations: LF, levator function; MRD1, marginal reflex distance 1.

SPSS version 24 (IBM), and GraphPad Prism version10 (Siemens).

Results

Data Characteristics

In the prospective validation phase in 2025, a total of 952 participants (1904 eyes) were eventually enrolled, including 459 males (48.2%) and 493 females (51.8%), with a mean age of 27.9 years (range, 3-86 years). Of the videos, 679 of 952 (71.3%) were acquired at the main center, and 273 of 952 (28.7%) at 4 subcenters. After automated quality control, 1862 eyes were included in the MRD1 analysis and 1830 eyes in the LF analysis; exclusions were mainly due to feature detection failure, substandard feature quality, or significant eyebrow movement. The mean MRD1 was 3.2 mm (range, −3.0 to 6.5 mm), and the mean LF was 7.6 mm (range, 0 to 16 mm). Among all participants, 206 patients (248 eyes) underwent surgery; the 3-month postoperative recurrence rate was 7.3% (95% CI, 4.4%-11.2%; 18/248). Detailed demographic and clinical characteristics are summarized in the **Table**. The CONSORT flow diagram is provided in eFigure 3 in Supplement 1.

MRD1 Evaluation Results

Quantitative assessment of MRD1 revealed a positive correlation between DRP and expert measurements (R = 0.88; 95% CI, 0.87-0.89; P < .001), with a MAE of 0.52 mm (95% CI, 0.50-0.54). Among 1862 eyes analyzed, measurement error was within 1.00 mm for 1600 eyes (85.9%) and within 0.50 mm for 1153 eyes (61.9%). A Bland-Altman plot demonstrated high agreement between methods, with a mean difference (bias) of 0.06 mm and 95% limits of agreement ranging from −1.31 mm to 1.43 mm. ICC was 0.93 (95% CI, 0.92-0.93) (**Figure 1**).

Based on MRD1 values, DRP automatically classified upper eyelid position as normal, droop from normal position, or ptosis. The AUC was 0.99 (95% CI, 0.99-1.00) for identifying droop from normal position (MRD1 <4 mm) and 0.97 (95% CI, 0.96-0.97) for diagnosing ptosis (MRD1 ≤2 mm); detailed diagnostic performance is summarized in eTable 2 in Supplement 1. The confusion matrix for 3-category classification is shown in Figure 1.

LF Evaluation Results

Quantitative assessment of LF also demonstrated a positive correlation between DRP and expert measurements (R = 0.92; 95% CI, 0.91-0.93; P < .001), with an MAE of 0.95 mm (95% CI, 0.91-0.98). Among 1830 eyes analyzed, measurement error was within 1.00 mm for 1155 eyes (63.1%) and within 0.50 mm for 888 eyes (48.5%). A Bland-Altman plot indicated high agreement between the 2 methods, with a mean difference of 0.01 mm between methods and 95% limits of agreement from −2.54 mm to 2.52 mm. The ICC was 0.95 (95% CI, 0.95-0.96) (**Figure 2**).

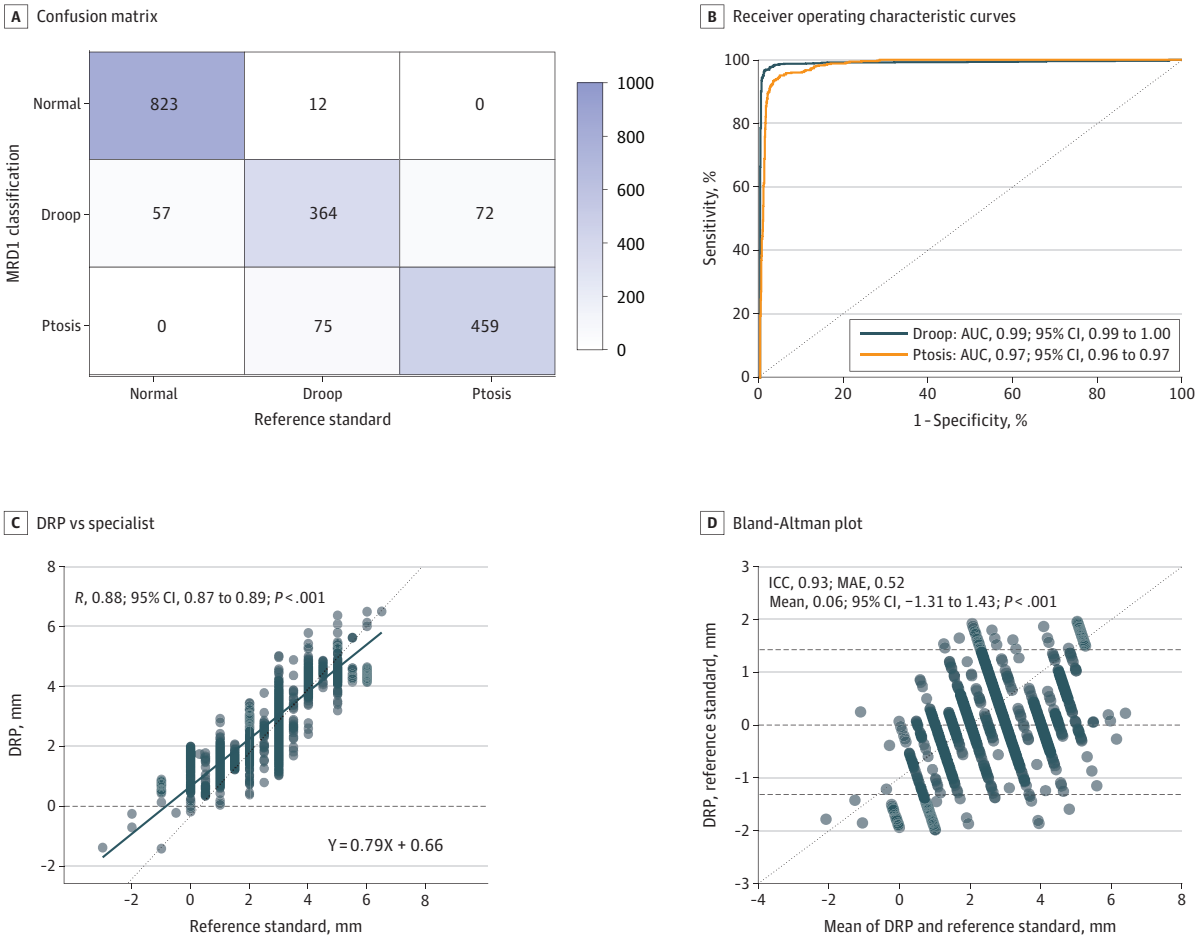
Based on LF values, DRP automatically categorized LF as good, fair, or severe. The AUC was 0.98 (95% CI, 0.97-0.98) for identifying abnormal muscle function (LF <7 mm) and 0.98 (95% CI, 0.97-0.99) for identifying severe muscle function (LF <4 mm); detailed diagnostic performance is summarized in eTable 3 in Supplement 1. The confusion matrix for the 3-category classification is shown in Figure 2. Additional operating-point analyses for both MRD1 and LF classifications, including sensitivity, specificity, positive predictive value, and negative predictive value at maximum Youden index thresholds, are provided in eTable 4 in Supplement 1.

Subgroup Analysis

To assess model robustness, subgroup analyses were performed across different ophthalmic centers (main center, subcenters), sexes (male, female), age groups (≤6 years, 7-59 years, and ≥60 years), and etiology (congenital, acquired, unknown). Across all 4 subgroups for MRD1 measurements, the maximum intergroup difference for assessment with an error 1 mm or less was 9.7% (ranging from 83.0% to 92.7%). For assessment with an error 0.5 mm or less, the maximum intergroup difference was 6.3% (ranging from 64.2% to 70.5%). For LF measurements, the maximum intergroup difference for assessment with an error 0.5 mm or less was 10.0% (ranging from 42.7% to 52.7%). For assessment with an error 0.5 mm or less, the maximum intergroup difference was 8.3% (ranging from 38.7% to 47.0%). Stacked bar charts illustrating error distributions are presented in eFigures 4 and 5 in Supplement 1.

Downloaded from jamanetwork.com by Tsinghua University user on 10/27/2026

Figure 1. Matrix, Curves, and Plots Showing the Performance of Quantitative Measurement of Marginal Reflex Distance 1 (MRD1) for 1862 Eyes



Confusion matrix (A) and receiver operating characteristic curves (B) for MRD1 classification (normal, droop from the normal position, ptosis). Scatterplot (C) and Bland-Altman plot (D) comparing Digital Ruler of Ptosis (DRP) predictions with specialist findings for MRD1 quantification. AUC indicates area under the curve; ICC, intraclass correlation coefficient; MAE, mean absolute error.

Comparison With 2D Methods

DRP's performance was benchmarked against prior studies using specialized instrumentation or highly standardized capture environments. A systematic search identified 11 relevant studies, 1 of which was excluded because its reference standard did not align with clinical guidelines.[17,18,29-37] The remaining studies reported an average sample size of 560 eyes, compared with 1904 eyes in our study. Among the 10 studies reporting MRD1 measurement, the average ICC of specialized instrumentation was 0.94 (comparable with the 0.93 achieved by DRP). For LF measurement, only 1 prior 2D-based study was identified, reporting an ICC of 0.69 (vs 0.95 in DRP) (**Figure 3**).

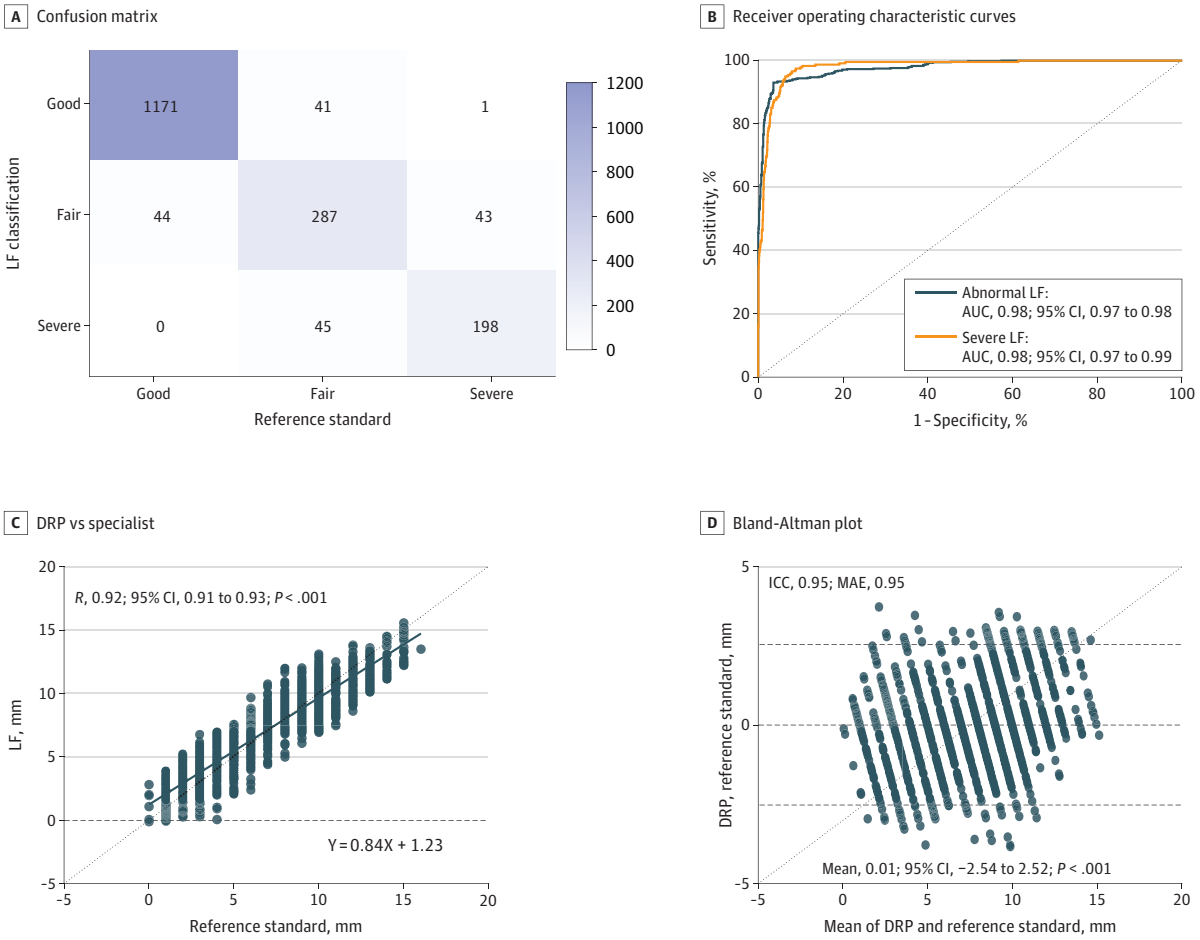
To further evaluate the advantage of our 3D reconstruction approach in a mobile application context, we reproduced a deep learning algorithm trained on 2D RGB appearance images using our dataset of 1904 eye examination videos. For the binary classification diagnosis of ptosis, DRP achieved a higher AUC (0.97; 95% CI, 0.96-0.99) than that the convolutional neural network method (0.89; 95% CI, 0.88-0.91) in the same prospective testing dataset (ΔAUC, 0.07; 95% CI, 0.06-0.09). Similarly, for binary classification of intervention strategy, DRP's AUC (0.98; 95% CI, 0.97-0.98) exceeded that of the convolutional neural network method (0.89; 95% CI, 0.87-0.90) (ΔAUC, 0.09; 95% CI, 0.08-0.11).

Discussion

In this prospective multicenter study, DRP used smartphone-based 3D reconstruction to quantify MRD1 and LF and classify ptosis-related surgical categories. DRP showed high agreement with specialist measurements, strong diagnostic performance, and consistent performance across sex, age, etiology, and clinical centers. Compared with the 2D baseline, DRP improved AUC for both ptosis diagnosis and intervention-strategy classification; the lower bounds of the 95% CIs for ΔAUC were 0.06 and 0.08, respectively, exceeding the lower end of the clinically relevant ΔAUC range described in diagnostic accuracy literature.[38-40] Decision curve analysis showed higher net benefit for DRP than the 2D baseline across clini-

Downloaded from jamanetwork.com by Tsinghua University user on 10/27/2026

Figure 2. Matrix, Curves, and Plots Showing the Performance of Quantitative Measurement of Levator Muscle Function (LF) for 1830 Eyes



Confusion matrix (A) and receiver operating characteristic curves (B) for LF classification (good, fair, and poor). Scatterplot (C) and Bland-Altman plot (D) comparing Digital Ruler of Ptosis (DRP) predictions with specialist findings for LF quantification. AUC indicates area under the curve; ICC, intraclass correlation coefficient; MAE, mean absolute error.

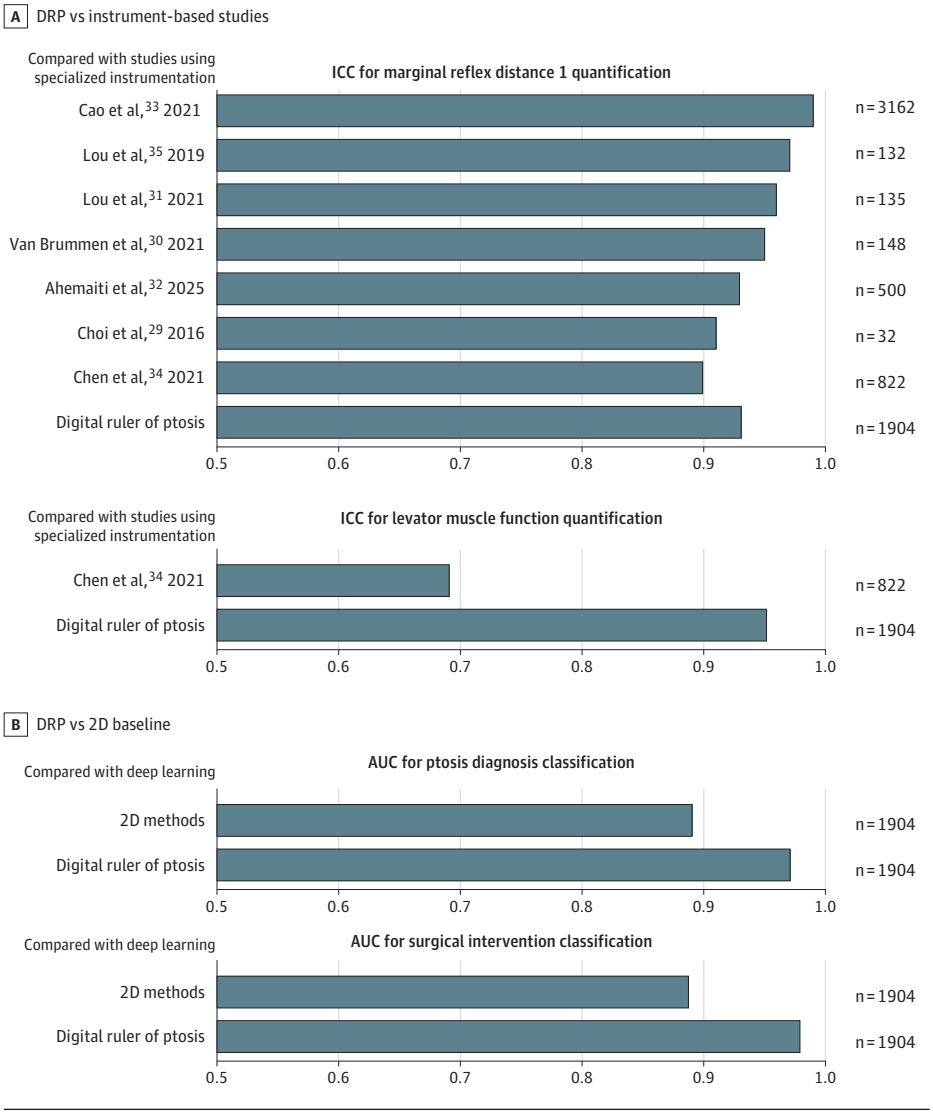
cally relevant threshold probabilities, supporting clinical utility beyond discrimination (eFigure 6 in Supplement 1).

Ptosis remains challenging to screen and monitor. As a visual development disorder affecting approximately 0.18% to 1.41% of children,[5,41] ptosis is frequently underdiagnosed because young children cannot reliably articulate symptoms, and caregivers may not recognize the condition. In older adults, upper eyelid skin laxity may obscure the true eyelid margin, causing overdiagnosis or severity overestimation.[42] Delayed intervention can lead to form-deprivation amblyopia and impaired ocular growth,[5,43] yet repeated severity and LF assessment depends on limited specialist resources.[44-46] Existing automated approaches remain limited. Regression-based methods can achieve accurate geometric measurements but typically require specialized hardware (eg, infrared or structured light cameras) and highly standardized capture protocols (eg, multiangle, with additional labeling),[16,17] limiting feasibility for at-home use. Smartphone RGB classification models offer lower hardware costs but generally lack the ability to provide highly precise quantitative measurements.[15] Their "black-box" nature also limits clinical interpretability,[47] hindering widespread clinical adoption. DRP bridges these gaps by measuring MRD1 and LF using only a smartphone with clinically interpretable outputs. This comparison was interpreted as contextual rather than head-to-head because cohorts, protocols, outcomes, and computational environments differed across studies. DRP required approximately 5 seconds and 0.257 watt hours per assessment, supporting feasibility for clinical or screening use.

Beyond accessibility, precise quantification carries direct clinical implications. Inaccurate MRD1 and LF measurements can increase missed diagnoses and misdiagnoses, lead to incorrect severity grading, and result in inappropriate surgical selection—all increasing the risk of suboptimal outcomes and repeat surgery. Notably, previous literature reports postoperative recurrence rates of 14.1% to 63.3%,[8-13] while in our surgical cohort, where standard-of-care planning was based on expert measurements, the 3-month recurrence rate was 7.3%. Although surgical planning in this study followed expert reference-standard assessment, DRP

Downloaded from jamanetwork.com by Tsinghua University user on 10/27/2026

Figure 3. Bar Chart Showing Comparison With Current Automatic Methods of Ptosis



A, Digital Rulers of Ptosis (DRP) vs prior published instrument-based studies for margin reflex distance 1 (MRD1) and levator muscle function (LF) quantifications. B, DRP vs a reproduced 2-dimensional (2D) deep-learning baseline on the same video dataset. AUC indicates area under the curve; CNN, convolutional neural network; ICC, intraclass correlation coefficient.

demonstrated high agreement with expert measurements, suggesting its potential to improve the reliability and reproducibility of preoperative MRD1 and LF quantification. However, whether DRP-guided planning could achieve comparable clinical outcomes or reduce recurrence and reoperation risks requires future evaluation. Prospective studies are being designed to explicitly compare clinical outcome between DRP-guided and conventional surgical planning and explore its potential value in determining the optimal surgical timing in children.

A further strength of this work is its explicit attention to real-world data quality in screening scenarios. Although medical AI moves toward practice,[48] many ocular-appearance models are developed on high-quality, laboratory-preprocessed images,[49] creating a gap with uncontrolled real-world acquisition that can lead to dataset bias and degraded deployment performance.[50] Unlike imaging modalities, such as optical coherence tomography or fundus photography, external ocular appearance capture is particularly sensitive to motion, pose variation, dynamic eyelid movement, eyebrow recruitment and occlusion, which may affect measurement reliability and diagnostic accuracy. We therefore developed a deep learning–based ocular-appearance quality control classifier that evaluates both clinically relevant confounders (eg, eyebrow-raising interference) and technical data quality failures (eg, abnormal iris detection, binocular data deviation), enabling automated real-time filtering and feedback. Integrating this module into DRP helps maintain analytic performance under practical acquisition conditions and support large-scale smartphone-based screening.

Limitations

This study has several limitations. First, postoperative recurrence rates were not compared between DRP-assisted and conventional care in a randomized clinical trial; they were assessed only at 3 months and pooled across heterogeneous

Downloaded from jamanetwork.com by Tsinghua University user on 10/27/2026

ptosis subtypes. Longer follow-up and controlled studies are needed. Second, manual specialist MRD1 and LF measurements served as the clinical reference standard but remain subject to interobserver variability. Third, video acquisition remains challenging in children and older adults because of limited cooperation and age-related skin laxity, respectively. Fourth, validation in ethnically diverse populations is needed to improve generalizability. Fifth, facial imaging-based diagnosis raises privacy concerns, which can be mitigated by local offline processing, deletion of raw videos after measurement, and digital mask technology that preserves ocular disease features while removing identifiable personal information.[22] Finally, DRP is currently not commercially available and requires regulatory approval before clinical commercial deployment. Current evidence supports DRP as an adjunct for standardized quantitative assessment in primary care, community screening, home monitoring after ptosis diagnosis, remote follow-up, and triage settings lacking specialist measurements.

Conclusions

This study found that consumer smartphones might support accurate, interpretable and scalable quantitative ptosis assessment. By enabling precise MRD1 and LF measurement with quality control for real-world screening, DRP may reduce missed and misdiagnosed cases and repeat-surgery risk through more consistent decision support. This framework may be extended to postoperative monitoring, including recurrence surveillance and remote longitudinal management of systemic conditions with ocular manifestations. This prospective clinical evaluation may serve as an example of moving beyond artificial intelligence tool development toward clinical evaluation. Following appropriate regulatory approval, randomized clinical trials should compare DRP-assisted and conventional care to determine whether DRP improves recurrence and other clinically relevant outcomes or achieves noninferior outcomes in the future.

ARTICLE INFORMATION

Accepted for Publication: July 26, 2026.

Published Online: September 24, 2026. doi:10.1001/jamaophthalmol.2026.4120

Open Access: This is an open access article distributed under the terms of the CC-BY-NC-ND License, which does not permit alteration or commercial use, including those for text and data mining, AI training, and similar technologies. © 2026 Yang Y et al. *JAMA Ophthalmology*.

Author Affiliations: Zhongshan Ophthalmic Centre, Sun Yat-sen University, WHO Collaborating Centre for Eye Care and Vision, State Key Laboratory of Ophthalmology, Guangdong Provincial Key Laboratory of Ophthalmology and Visual Science, Guangdong Provincial Clinical Research Center for Ocular Diseases, Guangzhou, Guangdong, China (Y. Yang, Wang, Ling, Zhao, Wu, Wen, Cui, Y. Li, Lai, Zeng, Chen, X. Li, Liang, Huang, Lin); School of Software and BNRist, Tsinghua University, Beijing, China (Han, Lyu, C. Li, Xu); Shenzhen Eye Hospital, Shenzhen Eye Medical Center, Southern Medical University, Shenzhen, Guangdong, China (Zhang, Ye, W. Yang); Department of Ophthalmology, The Second Affiliated Hospital of Fujian Medical University, Engineering Research Centre of Assistive Technology for Visual Impairment, Fujian Province University, Quanzhou, Fujian, China (Gao, Hu); Loudi Central Hospital, Loudi, Hunan, China (Gong, Zou); Department of Ophthalmology, First Affiliated Hospital of Guangzhou Medical University, Guangzhou, China (Meng); Cambridge Centre for Brain Repair and MRC Mitochondrial Biology Unit, Department of Clinical Neurosciences, University of Cambridge, Cambridge, United Kingdom (Yu-Wai-Man); Cambridge Eye Unit, Addenbrooke's Hospital, Cambridge University Hospitals, Cambridge, United Kingdom (Yu-Wai-Man); Moorfields Eye Hospital, London, United Kingdom (Yu-Wai-Man); UCL Institute of Ophthalmology, University College London, London, United Kingdom (Yu-Wai-Man); Hainan Eye Hospital and Key Laboratory of Ophthalmology, Zhongshan Ophthalmic Center, Sun Yat-sen University, Haikou, Hainan, China (Lin); Department of Genetics and Biomedical Informatics, Zhongshan School of Medicine, Institute for Frontier Interdisciplinary Research in Health Sciences and Technology, Sun Yat-sen University, Guangzhou, Guangdong, China (Lin).

Author Contributions: Drs Y. Yang and Zhao had full access to all of the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis. Drs Y. Yang, Han, Wang, and Ling contributed equally as co–first authors.
Concept and design: Y. Yang, Han, Wang, Ling, Lyu, Chen, Ye, W. Yang, Hu, X. Li, Liang, Xu, Lin.
Acquisition, analysis, or interpretation of data: Y. Yang, Wang, Ling, C. Li, Zhao, Wu, Wen, Cui, Y. Li, Lai, Zeng, Chen, Zhang, Gao, Gong, Zou, Meng, Huang, Yu-Wai-Man.
Drafting of the manuscript: Y. Yang, Han, Wang, Ling, Wen, Y. Li, Lin.
Critical review of the manuscript for important intellectual content: Y. Yang, Wang, Ling, Lyu, C. Li, Zhao, Wu, Cui, Lai, Zeng, Chen, Zhang, Ye, W. Yang, Gao, Hu, Gong, Zou, Meng, X. Li, Liang, Huang, Yu-Wai-Man, Xu.
Statistical analysis: Y. Yang, Han, Wang, Ling, Zhao, Y. Li, Chen, Zhang, Meng, Yu-Wai-Man.
Obtained funding: Y. Yang, Ling, Xu, Lin.
Administrative, technical, or material support: Y. Yang, Han, Ling, Lyu, Wen, Lai, Chen, W. Yang, Gao, Gong, Zou, X. Li, Liang, Huang, Xu.
Supervision: Ling, Ye, X. Li, Liang, Xu.

Conflict of Interest Disclosures: Dr Y. Yang reported a patent pending for a method, system, device, and medium for processing eye image data based on 3-dimensional reconstruction. Dr Han reported a patent pending for a method, system, device, and medium for processing eye image data based on 3-dimensional reconstruction. Dr Lin reported a patent pending for a method, system, device, and medium for processing eye image data based on 3-dimensional reconstruction. No other disclosures were reported.

Funding/Support: This study received support from the National Natural Science Fund of China (grants No. 82441003 [H.L.], 82301265 [Y.Y.], T2541076 [F.X.], T2541074 [F.X.]), Science and Technology Planning Project of Guangdong Province (grant No. 2023A1111120011 [H.L.]), Young Elite Scientists Sponsorship Program by CAST (2024QNRC001 [Y.Y.]), Science and Technology Planning Project of Guangzhou City (2025A03J3984 [H.L.]), Guangdong Basic Research Center of Excellence for Major Blinding Eye Diseases Prevention and Treatment, THUIBCS, Tsinghua University, and BLBCI, Beijing Municipal Education Commission.

Role of the Funder/Sponsor: The funders had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

Data Sharing Statement: See Supplement 2.

REFERENCES

1. Wright KW, Spiegel PH, Hengst T. *Pediatric Ophthalmology and Strabismus*. Springer Science & Business Media; 2013.

2. Bartley GB, Gorman CA. Diagnostic criteria for Graves' ophthalmopathy. *Am J Ophthalmol*. 1995; 119(6):792-795. doi:10.1016/S0002-9394(14)72787-4

3. Díaz-Manera J, Luna S, Roig C. Ocular ptosis: differential diagnosis and treatment. *Curr Opin Neurol*. 2018;31(5):618-627. doi:10.1097/WCO.0000000000000600

4. Griepentrog GJ, Diehl NN, Mohney BG. Incidence and demographics of childhood ptosis. *Ophthalmology*. 2011;118(6):1180-1183. doi:10.1016/j.ophtha.2010.10.026

5. Pavone P, Cho SY, Praticò AD, Falsaperla R, Ruggieri M, Jin DK. Ptosis in childhood: a clinical sign of several disorders: case series reports and literature review. *Medicine (Baltimore)*. 2018;97 (36):e12124. doi:10.1097/MD.0000000000012124

6. Anderson RL, Baumgartner SA. Amblyopia in ptosis. *Arch Ophthalmol*. 1980;98(6):1068-1069. doi:10.1001/archopht.1980.01020031058009

7. Lehnerer S, Jacobi J, Schilling R, et al. Burden of disease in myasthenia gravis: taking the patient's

Downloaded from jamanetwork.com by Tsinghua University user on 10/27/2026

perspective. *J Neurol*. 2022;269(6):3050-3063. doi:10.1007/s00415-021-10891-1

8. Ho YF, Wu SY, Tsai YJ. Factors associated with surgical outcomes in congenital ptosis: a 10-year study of 319 cases. *Am J Ophthalmol*. 2017;175:173-182. doi:10.1016/j.ajo.2016.12.013

9. Ben Simon GJ, Macedo AA, Schwarcz RM, Wang DY, McCann JD, Goldberg RA. Frontalis suspension for upper eyelid ptosis: evaluation of different surgical designs and suture material. *Am J Ophthalmol*. 2005;140(5):877-885. doi:10.1016/j.ajo.2005.05.031

10. Lee MJ, Oh JY, Choung HK, Kim NJ, Sung MS, Khwarg SI. Frontalis sling operation using silicone rod compared with preserved fascia lata for congenital ptosis a three-year follow-up study. *Ophthalmology*. 2009;116(1):123-129. doi:10.1016/j.ophtha.2008.08.049

11. Hayashi K, Katori N, Kasai K, Kamisasanuki T, Kokubo K, Ohno-Matsui K. Comparison of nylon monofilament suture and polytetrafluoroethylene sheet for frontalis suspension surgery in eyes with congenital ptosis. *Am J Ophthalmol*. 2013;155(4):654-663.e1. doi:10.1016/j.ajo.2012.10.022

12. Woo KI, Kim YD, Kim YH. Surgical treatment of severe congenital ptosis in patients younger than two years of age using preserved fascia lata. *Am J Ophthalmol*. 2014;157(6):1221-1226.e1. doi:10.1016/j.ajo.2014.02.041

13. Chow K, Deva N, Ng SG. Prolene frontalis suspension in paediatric ptosis. *Eye (Lond)*. 2011;25(6):735-739. doi:10.1038/eye.2011.32

14. Hamet P, Tremblay J. Artificial intelligence in medicine. *Metabolism*. 2017;69:S36-S40. doi:10.1016/j.metabol.2017.01.011

15. Shu Q, Pang J, Liu Z, et al. Artificial intelligence for early detection of pediatric eye diseases using mobile photos. *JAMA Netw Open*. 2024;7(8):e2425124-e2425124. doi:10.1001/jamanetworkopen.2024.25124

16. Song X, Tong W, Lei C, et al. A clinical decision model based on machine learning for ptosis. *BMC Ophthalmol*. 2021;21(1):169. doi:10.1186/s12886-021-01923-5

17. Song B, Kwon H, Kim S, Ha Y, Oh SH, Song SH. Introduction of deep learning-based infrared image analysis to marginal reflex distance1 measurement method to simultaneously capture images and compute results: clinical validation study. *J Clin Med*. 2023;12(23):7466. doi:10.3390/jcm12237466

18. Bodnar ZM, Neimkin M, Holds JB. Automated ptosis measurements from facial photographs. *JAMA Ophthalmol*. 2016;134(2):146-150. doi:10.1001/jamaophthalmol.2015.4614

19. Mason J, Dave R, Chatterjee P, Graham-Allen I, Esterline A, Roy K. An investigation of biometric authentication in the healthcare environment. *Array (N Y)*. 2020;8:100042. doi:10.1016/j.array.2020.100042

20. Wen Q, Xu F, Yong JH. Real-time 3D eye performance reconstruction for RGBD cameras. *IEEE Trans Vis Comput Graph*. 2017;23(12):2586-2598. doi:10.1109/TVCG.2016.2641442

21. Ploumpis S, Ververas E, Sullivan EO, et al. Towards a complete 3D morphable model of the human head. *IEEE Trans Pattern Anal Mach Intell*. 2021;43(11):4142-4160. doi:10.1109/TPAMI.2020.2991150

22. Yang Y, Lyu J, Wang R, et al. A digital mask to safeguard patient privacy. *Nat Med*. 2022;28(9):1883-1892. doi:10.1038/s41591-022-01966-1

23. Wang R, Lyu J, Yang Y, et al. A smartphone-based digital ruler to automatically measure strabismus in ophthalmologist-level: a prospective, multicenter cohort study. *NEJM AI*. 2025;2(11):AIoa2401205. doi:10.1056/AIoa2401205

24. Small RG, Sabates NR, Burrows D. The measurement and definition of ptosis. *Ophthalmic Plast Reconstr Surg*. 1989;5(3):171-175. doi:10.1097/00002341-198909000-00003

25. Paik JS, Kim SA, Park SH, Yang SW. Refractive error characteristics in patients with congenital blepharoptosis before and after ptosis repair surgery. *BMC Ophthalmol*. 2016;16(1):177. doi:10.1186/s12886-016-0351-9

26. Vichare NV. Retracted: Prevalence of refractive errors and corneal topographic changes in unilateral congenital ptosis vs the fellow eye: a clinical study. *Int J Res Med Sci*. 2019;7(2):442-446. doi:10.18203/2320-6012.ijrms20190350

27. Expert Group for the Consensus on the Diagnosis and Treatment of Blepharoptosis. Expert consensus on the diagnosis and treatment of blepharoptosis [in Chinese]. *Zhonghua Yi Xue Za Zhi*. 2017;97(6):406-411. doi:10.3760/cma.j.issn.0376-2491.2017.06.002

28. Patel SM, Linberg JV, Sivak-Callcott JA, Gunel E. Modified tarsal resection operation for congenital ptosis with fair levator function. *Ophthalmic Plast Reconstr Surg*. 2008;24(1):1-6. doi:10.1097/IOP.0b013e31815e72a3

29. Choi CJ, Lefebvre DR, Yoon MK. Validation of the facial assessment by computer evaluation (FACE) program for software-aided eyelid measurements. *Orbit*. 2016;35(3):117-120. doi:10.3109/01676830.2016.1139595

30. Van Brummen A, Owen JP, Spaide T, et al. PeriorbitAI: artificial intelligence automation of eyelid and periorbital measurements. *Am J Ophthalmol*. 2021;230:285-296. doi:10.1016/j.ajo.2021.05.007

31. Lou L, Cao J, Wang Y, et al. Deep learning-based image analysis for automated measurement of eyelid morphology before and after blepharoptosis surgery. *Ann Med*. 2021;53(1):2278-2285. doi:10.1080/07853890.2021.2009127

32. Ahemaiti A, Chen S, Lei S, Liu L, Zhao M. Deep learning based adaptive and automatic measurement of palpebral margin in eyelid morphology. *Sci Rep*. 2025;15(1):13914. doi:10.1038/s41598-025-93975-6

33. Cao J, Lou L, You K, et al. A novel automatic morphologic analysis of eyelids based on deep learning methods. *Curr Eye Res*. 2021;46(10):1495-1502. doi:10.1080/02713683.2021.1908569

34. Chen HC, Tzeng SS, Hsiao YC, Chen RF, Hung EC, Lee OK. Smartphone-based artificial intelligence–assisted prediction for eyelid measurements: algorithm development and observational validation study. *JMIR Mhealth Uhealth*. 2021;9(10):e32444. doi:10.2196/32444

35. Lou L, Yang L, Ye X, et al. A novel approach for automated eyelid measurements in blepharoptosis using digital image analysis. *Curr Eye Res*. 2019;44(10):1075-1079. doi:10.1080/02713683.2019.1619779

36. Rana K, Beecher M, Caltabiano C, et al. Artificial intelligence to automate assessment of ocular and periocular measurements. *Eur J Ophthalmol*. 2025;35(1):346-351. doi:10.1177/11206721241249773

37. Chun YS, Park HH, Park IK, Moon NJ, Park SJ, Lee JK. Topographic analysis of eyelid position using digital image processing software. *Acta Ophthalmol*. 2017;95(7):e625-e632. doi:10.1111/aos.13437

38. Guyatt G, Jaeschke R, Heddle N, Cook D, Shannon H, Walter S. Basic statistics for clinicians: 2. Interpreting study results: confidence intervals. *CMAJ*. 1995;152(2):169-173.

39. Emir B, Can FE, Kaymaz E, et al. Sample size and power analysis for ROC AUC differences in diagnostic tests: a methodological evaluation of the Obuchowski-McClish and Hanley-McNeil methods. *BMC Med Res Methodol*. 2026;26(1):43. doi:10.1186/s12874-026-02768-6

40. Vickers AJ, Elkin EB. Decision curve analysis: a novel method for evaluating prediction models. *Med Decis Making*. 2006;26(6):565-574. doi:10.1177/0272989X06295361

41. SooHoo JR, Davies BW, Allard FD, Durairaj VD. Congenital ptosis. *Surv Ophthalmol*. 2014;59(5):483-492. doi:10.1016/j.survophthal.2014.01.005

42. Finsterer J. Ptosis: causes, presentation, and management. *Aesthetic Plast Surg*. 2003;27(3):193-204. doi:10.1007/s00266-003-0127-5

43. Wang Y, Xu Y, Liu X, Lou L, Ye J. Amblyopia, strabismus and refractive errors in congenital ptosis: a systematic review and meta-analysis. *Sci Rep*. 2018;8(1):8320. doi:10.1038/s41598-018-26671-3

44. Resnikoff S, Felch W, Gauthier TM, Spivey B. The number of ophthalmologists in practice and training worldwide: a growing gap despite more than 200,000 practitioners. *Br J Ophthalmol*. 2012;96(6):783-787. doi:10.1136/bjophthalmol-2011-301378

45. Adashi EY, Geiger HJ, Fine MD. Health care reform and primary care: the growing importance of the community health center. *N Engl J Med*. 2010;362(22):2047-2050. doi:10.1056/NEJMp1003729

46. Boboridis K, Assi A, Indar A, Bunce C, Tyers AG. Repeatability and reproducibility of upper eyelid measurements. *Br J Ophthalmol*. 2001;85(1):99-101. doi:10.1136/bjo.85.1.99

47. Ribeiro MT, Singh S, Guestrin C. "Why should I trust you?" explaining the predictions of any classifier. *arXiv*. Preprint posted online February 16, 2016. doi:10.48550/arXiv.1602.04938

48. Bajwa J, Munir U, Nori A, Williams B. Artificial intelligence in healthcare: transforming the practice of medicine. *Future Healthc J*. 2021;8(2):e188-e194. doi:10.7861/fhj.2021-0095

49. Zhang Z, Wang Y, Zhang H, et al. Artificial intelligence-assisted diagnosis of ocular surface diseases. *Front Cell Dev Biol*. 2023;11:1133680. doi:10.3389/fcell.2023.1133680

50. Ji Y, Liu S, Hong X, et al. Advances in artificial intelligence applications for ocular surface diseases diagnosis. *Front Cell Dev Biol*. 2022;10:1107689. doi:10.3389/fcell.2022.1107689

Downloaded from jamanetwork.com by Tsinghua University user on 10/27/2026